

Структура директории, где будет твориться темная магия:

```
total 20
drwxr-xr-x 5 root root 4096 Jun 10 08:55 .
drwx----- 5 root root 4096 Jun 10 08:55 ..
drwxr-xr-x 2 root root 4096 Jun 10 08:55 output
drwxr-xr-x 2 root root 4096 Jun 10 08:55 rootfs
drwxr-xr-x 2 root root 4096 Jun 10 08:55 src
root@kivvi-VirtualBox:~/arm_lab#
```

Сборка zlib

```
root@kivvi-VirtualBox:~/arm_lab/src/zlib-1.3.2# CC=arm-linux-gnueabi-gcc ./configure --prefix=/usr
c/zlib-1.3.2# make -j$(nproc)
-D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -D_MINGZIPSH -D_MINGZIPSH
c/zlib-1.3.2# make DESTDIR=$HOME/arm_lab/rootfs install
/libz.a
```

Сборка openssl

```
root@kivvi-VirtualBox:~/arm_lab/src/openssl-3.0.16# ./Configure linux-armv4 --cross-compile-prefix=arm-linux-gnueabi-gcc --prefix=/usr shared
Configuring OpenSSL version 3.0.16 for target linux-armv4
Using os-specific seed configuration
Created configdata.pm
Running configdata.pm
Created Makefile.in
Created Makefile
Created include/openssl/configuration.h

*****
***
*** OpenSSL has been successfully configured
***
*** If you encounter a problem while building, please open an
*** issue on GitHub <https://github.com/openssl/openssl/issues>
*** and include the output from the following command:
***
***     perl configdata.pm --dump
***
*** (If you are new to OpenSSL, you might want to consult the
*** 'Troubleshooting' section in the INSTALL.md file first)
***
*****
root@kivvi-VirtualBox:~/arm_lab/src/openssl-3.0.16# make -j$(nproc)
```

openssh

```
root@kivvi-VirtualBox:~/arm_lab/src/openssh-portable# ./configure --host=arm-linux-gnueabi-gcc --prefix=/usr --sysconfdir=/etc/ssh --with-zlib=$HOME/arm_lab/rootfs/usr --with-ssl-dir=$HOME/arm_lab/rootfs/usr --disable-strip
/openssh-portable# make -j$(nproc)
/openssh-portable# make DESTDIR=$HOME/arm_lab/rootfs install
```

Make выдаст ошибку 225, но она связана с попыткой запустить ssh для проверки. Разумеется, запустить ssh под arm на x86 не получается, так что ошибку глубоко игнорируем.

Копирование ключевых либ (как я понял, libc и не только) уже под arm:

```
root@kivvi-VirtualBox:~# cp -a /usr/arm-linux-gnueabi/lib/* ~/arm_lab/rootfs/lib/
root@kivvi-VirtualBox:~# cp -a /usr/lib/arm-linux-gnueabi/* ~/arm_lab/rootfs/usr/lib/
root@kivvi-VirtualBox:~#
```

Создание root'a:

```
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc# cat passwd
root:x:0:0:root:/root:/bin/sh
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc# cat group
root:x:0:
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc# cat shadow
root::10933:0:99999:7:::
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc# chmod +600 shadow
```

конфига для ssh

```
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc# cat ssh/sshd_config
Port 22
PermitRootLogin yes
PasswordAuthentication yes
UsePAM no
Subsystem sftp /usr/libexec/sftp-server
root@kivvi-VirtualBox:~/arm_lab/rootfs/etc#
```

Файл init

```
#!/bin/sh

mount -t proc proc /proc
mount -t sysfs sysfs /sys
mount -t devtmpfs devtmpfs /dev

mkdir -p /var/run
mkdir -p /run/sshd

echo "ARM Linux booted"

exec /bin/sh

chmod +x ~/arm_lab/rootfs/init
```

Создание архива

```
rootfs# find . -print0 | cpio --null -ov --format=newc | gzip -9 > ../output/initramfs.cpio.gz
```

Запуск виртуалки:

```
root@kivvi-VirtualBox:~/arm_lab# QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -m 256M -kernel zImage -dtb vexpress-v2p-c  
a9.dtb -initrd output/initramfs.cpio.gz -append "console=ttyAMA0 rdinit=/init" -nographic
```

Итог:

```
~ # whoami  
root  
~ # ssh  
[ 70.967768] random: crng init done  
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface] [-b bind_address]  
        [-c cipher_spec] [-D [bind_address:]port] [-E log_file]  
        [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]  
        [-J destination] [-L address] [-l login_name] [-m mac_spec]  
        [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]  
        [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]  
        destination [command [argument ...]]  
ssh [-Q query_option]  
~ #
```

```
~ # ssh-keygen -A  
ssh-keygen: generating new host keys: RSA ECDSA ED25519
```